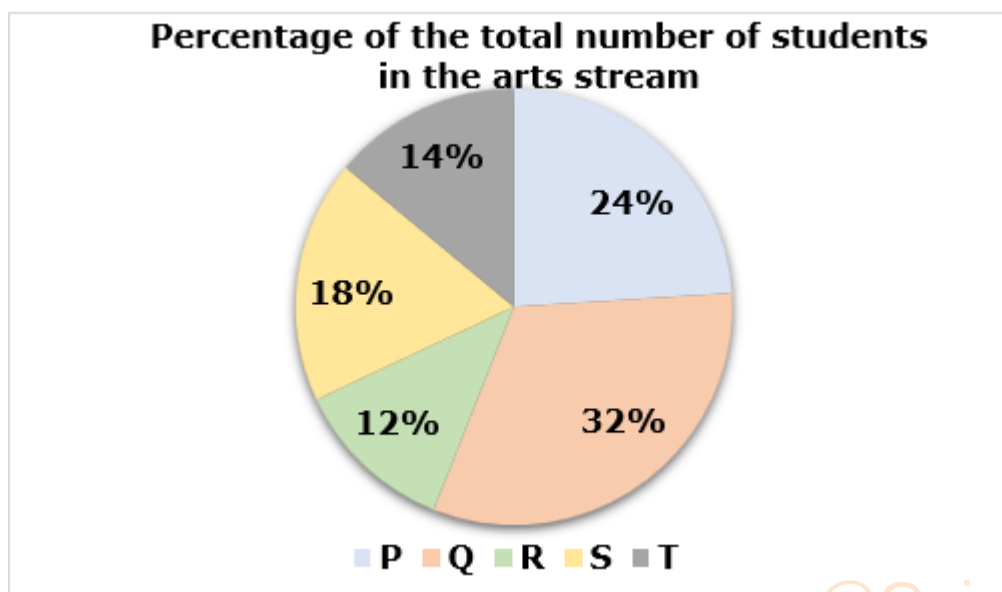
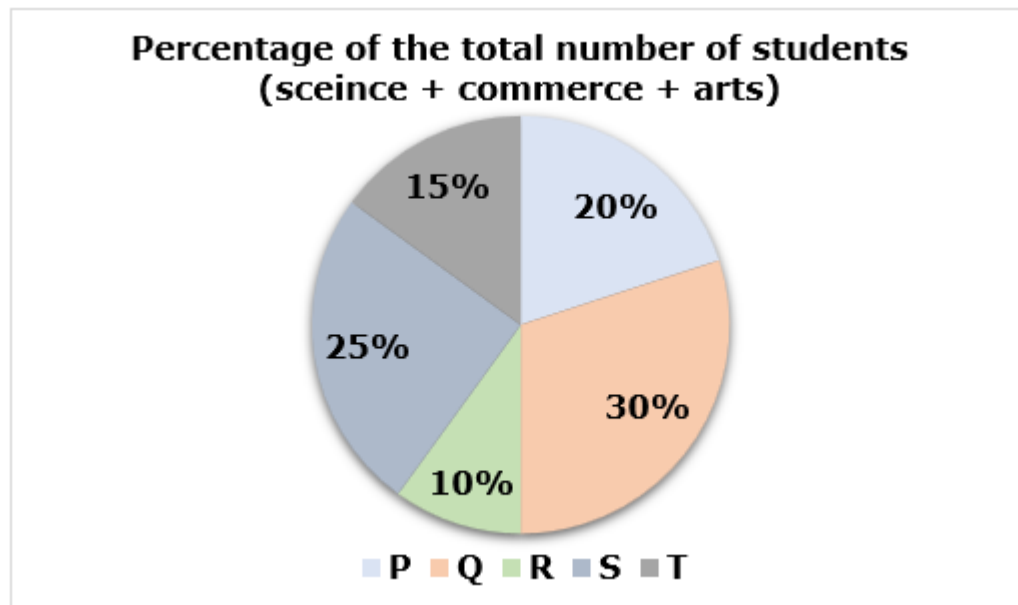


1. Questions

Read the following information carefully and answer the questions based on it.

The first pie chart given below shows the percentage distribution of the total number of students (science + commerce + arts) in five schools P, Q, R, S and T. The second pie chart given below shows the percentage distribution of the total number of students in the arts stream in each school.



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Note:

- I).** The total number of students = the number of students in the science stream+ the number of students in the commerce stream+ the number of students in the arts stream
- II).** In each school, the number of students in the science stream is more than that in the commerce stream.
- III).** $\frac{1}{5}^{\text{th}}$ of the total number of students in all five schools together are in the arts stream.
- IV).** In school S, the ratio of the number of students in the science stream to that in the arts is 5:1.
- V).** The difference between the number of students in the science stream and the commerce stream in

schools P, Q, R, S, and T is 152, 128, 28, 292, and 60 respectively. (science > commerce)

In school R, if $\frac{1}{3}^{\text{rd}}$ of the total number of students in the science stream are girls, which is 16 more than the number of girls in the commerce stream, find the difference between the number of boys in the science and the commerce stream.

- a. 17
- b. 24
- c. 14
- d. 21
- e. None of these

2. Questions

In school Q, out of the total number of students, some students appeared, while the rest did not appear in the examination, which is equal to the number of students who failed in the examination. If the ratio of the number of students who passed and failed the examination is 4:1, then find what percent of the total number of students did not appear in the examination from school Q?

- a. 33.33%
- b. 16.67%
- c. 37.50%
- d. Cannot be determined
- e. None of these

3. Questions

In school P, the total number of teachers (males + females) is [X] and the ratio of the number of male and female teachers is 19:9. If the number of male teachers is $\frac{1}{6}^{\text{th}}$ of the total number of students in science in school P, then find the value of [X-15].

- a. 41
- b. 56
- c. 33
- d. 28
- e. 77

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4. Questions

The total number of students in another school U is $\frac{1}{3}^{\text{rd}}$ of that in school T and the ratio of the number of students in the science and commerce stream is 8:5. If the number of students in the arts stream in school U is 22, then find the number of students in the commerce stream in school U.

- a. 48
- b. 64
- c. 56
- d. 30
- e. 36

5. Questions

Which of the following is/are definitely true?

I: Only in schools P and S, the number of students in the commerce stream is less than in the arts stream.

II: In school Q, $\frac{1}{3}^{\text{rd}}$ of the total number of students are in the science stream.

III: The difference between the number of students in the commerce stream in schools R and T is 30.

- a. Both I and II
- b. Both I and III
- c. Only II
- d. All I, II and III
- e. Both II and III

6. Questions

In school R, if the average age of each boy and each girl is 16 years and 14 years respectively and the total age of all students (boys + girls) is 3040 years, then find the number of girls in school R.

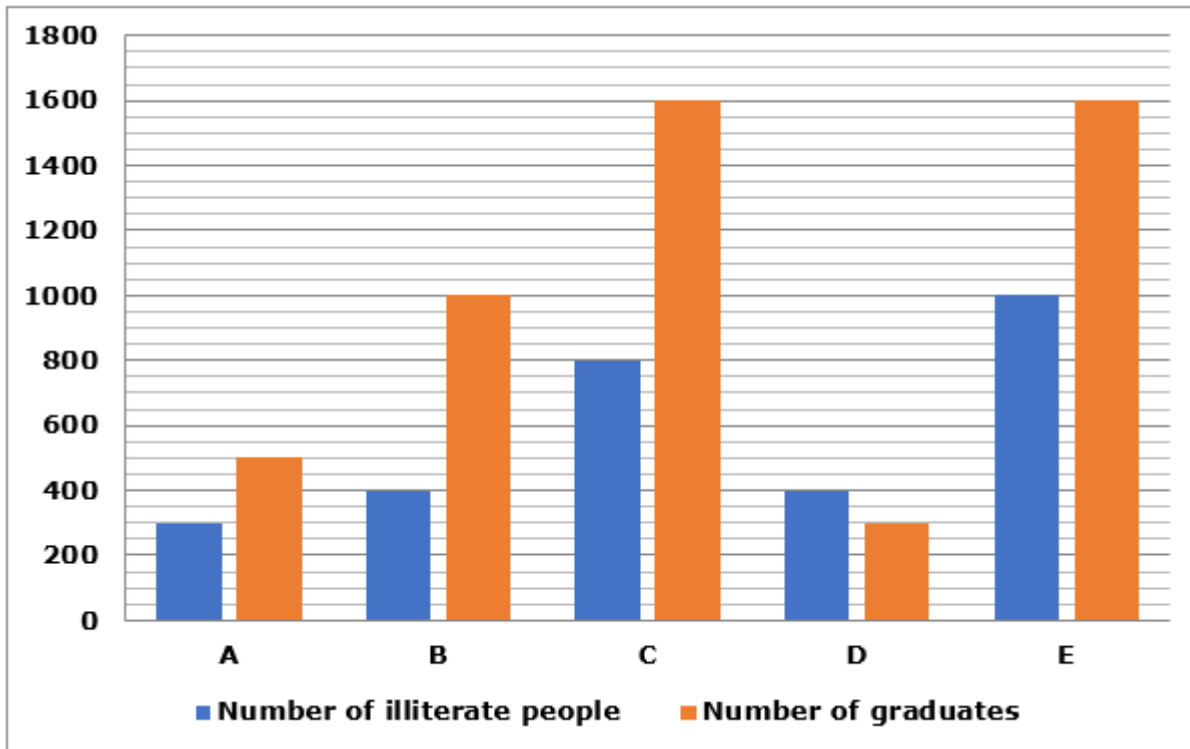
- a. 80
- b. 120
- c. 75
- d. 105
- e. 90

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7. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the number of illiterate people in five villages A, B, C, D, and E. It also shows the number of graduates in each village.



Note:

- I). The total number of people = the number of literate people + the number of illiterate people
- II). The total number of literate people = the number of graduates + the number of post-graduates
- III). Only literate people can be graduates and post-graduates.
- IV). The ratio of the total number of people and the number of post-graduates in villages A, B, C, D, and E is 3:1, 10:3, 4:1, 10:3, and 15:2 respectively.

In village A, if $\frac{2}{3}^{\text{rd}}$ of the total number of literate people are males and the ratio of the number of males and females who are post-graduates is 5:3, then how much percent of the total number of graduates in village A are females?

- a. 12%
- b. 24%
- c. 30%
- d. Cannot be determined
- e. None of these

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8. Questions

In village B, if 33.33% of the number of post-graduates are unemployed, which is 80 more than the number of graduates who are unemployed, then find the ratio of the number of graduates who are employed to that of post-graduates in village B.

- a. 11:5
- b. 12:1

- c. 13:8
- d. 16:9
- e. 7:8

9. Questions

The total number of literate people in another village F is 600 more than in village C. In village F, if the ratio of the number of graduates and post-graduates is 3:2 and the number of post-graduates is 20% more than the number of illiterate people, then find the total number of people in village F.

- a. 3200
- b. 4000
- c. 5600
- d. 2400
- e. 4800

10. Questions

In village D, if 30% of the total number of people are females and the ratio of the number of males and females who are literate is 9:1, then find the number of males who are illiterate in village D.

- a. 240
- b. 360
- c. 120
- d. Cannot be determined
- e. 160

11. Questions

Determine the correct relation among [X], [Y], and [Z].

X = The average number of literate people in villages A and D together.

Y = The difference between the total number of people in villages B and C.

Z = The total number of post-graduates in villages A and E.

- a. $X > Y < Z$
- b. $X < Y > Z$
- c. $X > Y > Z$
- d. $X < Y < Z$
- e. $X = Y = Z$

12. Questions

In village C, out of the total number of post-graduates, some are in private colleges, while the rest are in government colleges. If the number of post-graduates in government colleges are $[N]$, which is $[M]$ less than the number of post-graduates in private colleges, then find the value of $[2M-N]$. ($N-M = 100$)

- a. 200
- b. 400
- c. 100
- d. 300
- e. 90

13. Questions

Read the following information carefully and answer the questions based on it.

Five persons P, Q, R, S, and T are employed to work on a certain project.

The time taken by P and Q together to complete the whole work is 12 days. R, S, and T working together can complete the whole work in $(150/31)$ days. P and S together can complete 50% of the total work in $(10/3)$ days. Working with half of his actual efficiency, T can complete the whole work in 30 days. The time taken by P and R together to complete the whole work is $(100/9)$ days.

Note:

All five persons are working on the same project but the individual efficiency of each person may not be the same.

If P reduced his efficiency by $[X]\%$ and Q increased his efficiency by $[2X+10]\%$, then the time taken by P and Q together to complete the whole work will be $(100/9)$ days. Find the value of $[X/5]$.

- a. 4
- b. 12
- c. 8
- d. 20
- e. 6

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14. Questions

If the time taken by R and U together to complete the whole work is $(50/3)$ days, whereas U and V together can complete the same work in $(300/11)$ days, then V will complete the same work alone in ____ days.

What should come in place of the question mark?

- a. 50

- b. 40
- c. 30
- d. 60
- e. 64

15. Questions

If S and T started working on another project in such a way that S and T together worked for [N] days, then S left the work and T alone completed the remaining 50% of the total work in [2N+2] days, then find the total work done by S and T.

Note: the individual efficiency of S and T is unchanged.

- a. 200 units
- b. 300 units
- c. 400 units
- d. 500 units
- e. None of these

16. Questions

The following question has two equations as equation I and equation II. You have to solve both equations and determine the relationship between both equations and give an answer accordingly.

I). $5X^2 - 26X + 32 = 0$

II). $3Y^2 - 31Y + 56 = 0$

- a. $X > Y$
- b. $X < Y$
- c. $X \geq Y$
- d. $X \leq Y$
- e. $X = Y$, or relationship cannot be established

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17. Questions

I). $4A^2 - 29A + 51 = 0$

II). $2B^2 - 27B + 90 = 0$

- a. $A > B$
- b. $A < B$
- c. $A \geq B$

d. $A \leq B$

e. $A = B$ or relationship cannot be established

18. Questions

I). $P^2 + 4P - 480 = 0$

II). $Q^2 - 17Q - 390 = 0$

a. $P > Q$

b. $P < Q$

c. $P \geq Q$

d. $P \leq Q$

e. $P = Q$ or the relationship cannot be established

19. Questions

I). $M^2 - 42M + 341 = 0$

II). $N^2 - 11N - 152 = 0$

a. $M > N$

b. $M < N$

c. $M \geq N$

d. $M \leq N$

e. $M = N$ or relationship cannot be established

20. Questions

I). $6K^2 + 23K - 13 = 0$

II). $8L^2 - 29L + 21 = 0$

a. $K > L$

b. $K < L$

c. $K \geq L$

d. $K \leq L$

e. $K = L$ or relationship cannot be established

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Explanations:

1. Questions

Let the total number of students in all five schools together be 500k.

So, the total number of students in the arts stream in all five schools together = $(1/5) * 500k = 100k$.

Now,

The total number of students in school S = $(25/100) * 500k = 125k$

The total number of students in the arts stream in school S = $(18/100) * 100k = 18k$

So, the sum of the number of students in the science and the commerce stream together in school S = $(125k - 18k) = 107k$

The number of students in the science stream = $(18k * 5) = 90k$

So, the number of students in the commerce stream = $(107k - 90k) = 17k$

Now,

$$90k - 17k = 292$$

$$73k = 292$$

$$k = 4$$

So, the total number of students in all five schools together = $500k = (500 * 4) = 2000$

And, the total number of students in the arts stream in all five schools together = $100k = (100 * 4) = 400$

In School P,

The total number of students = $(20/100) * 2000 = 400$

The number of students in the arts stream = $(24/100) * 400 = 96$

The sum of the number of students in the science and commerce stream together = $(400 - 96) = 304$

Let the number of students in the science stream be a.

So, the number of students in the commerce stream = $(304 - a)$

$$\text{ATD, } a - (304 - a) = 152$$

$$2a = 456$$

$$a = 228$$

The number of students in the science stream = 228

The number of students in the commerce stream = $304 - 228 = 76$

In School Q

The total number of students = $(30/100) * 2000 = 600$

The number of students in the arts stream = $(32/100) * 600 = 128$

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The sum of the number of students in the science and commerce stream together = $(600 - 128) = 472$

Let the number of students in the science stream be b .

So, the number of students in the commerce stream = $(472 - b)$

ATD, $b - (472 - b) = 128$

$$2b = 600$$

$$b = 300$$

The number of students in the science stream = 300

The number of students in the commerce stream = $472 - 300 = 172$

In School R

The total number of students = $(10/100) * 2000 = 200$

The number of students in the arts stream = $(12/100) * 400 = 48$

The sum of the number of students in the science and commerce stream together = $(200 - 48) = 152$

Let the number of students in the science stream be c .

So, the number of students in the commerce stream = $(152 - c)$

ATD, $c - (152 - c) = 28$

$$2c = 180$$

$$c = 90$$

The number of students in the science stream = 90

The number of students in the commerce stream = $152 - 90 = 62$

In School S

The total number of students = $(25/100) * 2000 = 500$

The number of students in the arts stream = $(18/100) * 400 = 72$

The sum of the number of students in the science and commerce stream together = $(500 - 72) = 428$

Let the number of students in the science stream be 'd'.

So, the number of students in the commerce stream = $(428 - d)$

ATD, $d - (428 - d) = 292$

$$2d = 720$$

$$d = 360$$

The number of students in the science stream = 360

The number of students in the commerce stream = $428 - 360 = 68$

In School T

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The total number of students = $(15/100) * 2000 = 300$

The number of students in arts = $(14/100) * 400 = 56$

The sum of the number of students in science and commerce = $(300-56) = 244$

Let the number of students in the science stream be 'e'.

So, the number of students in the commerce stream = $(244 - e)$

ATD, $e - (244 - e) = 60$

$2e = 304$

$e = 152$

The number of students in the science stream = 152

The number of students in the commerce stream = $244 - 152 = 92$

School	The total number of students	The number of students in the science stream	The number of students in the commerce stream	The number of students in the arts stream
P	400	228	76	96
Q	600	300	172	128
R	200	90	62	48
S	500	360	68	72
T	300	152	92	56
Total	2000	1130	470	400

Answer: E

Now, according to the question,

In School R,

The total number of students in the science stream = 90

The number of girls in the science stream = $(1/3) * 90 = 30$

The number of boys in the science stream = $(90-30) = 60$

The total number of students in the commerce stream = 62

The number of girls in the commerce stream = $(30-16) = 14$

The number of boys in the commerce stream = $(62-14) = 48$

Therefore, the required difference = $(60-48) = 12$

Hence, the correct answer is option E.

2. Questions

Let the total number of students in all five schools together be 500k.

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So, the total number of students in the arts stream in all five schools together = $(1/5) * 500k = 100k$.

Now,

The total number of students in school S = $(25/100) * 500k = 125k$

The total number of students in the arts stream in school S = $(18/100) * 100k = 18k$

So, the sum of the number of students in the science and the commerce stream together in school S = $(125k - 18k) = 107k$

The number of students in the science stream = $(18k * 5) = 90k$

So, the number of students in the commerce stream = $(107k - 90k) = 17k$

Now,

$$90k - 17k = 292$$

$$73k = 292$$

$$k = 4$$

So, the total number of students in all five schools together = $500k = (500 * 4) = 2000$

And, the total number of students in the arts stream in all five schools together = $100k = (100 * 4) = 400$

In School P,

The total number of students = $(20/100) * 2000 = 400$

The number of students in the arts stream = $(24/100) * 400 = 96$

The sum of the number of students in the science and commerce stream together = $(400 - 96) = 304$

Let the number of students in the science stream be a.

So, the number of students in the commerce stream = $(304 - a)$

$$\text{ATD, } a - (304 - a) = 152$$

$$2a = 456$$

$$a = 228$$

The number of students in the science stream = 228

The number of students in the commerce stream = $304 - 228 = 76$

In School Q

The total number of students = $(30/100) * 2000 = 600$

The number of students in the arts stream = $(32/100) * 400 = 128$

The sum of the number of students in the science and commerce stream together = $(600 - 128) = 472$

Let the number of students in the science stream be b.

So, the number of students in the commerce stream = $(472 - b)$

$$\text{ATD, } b - (472 - b) = 128$$

$$2b = 600$$

$$b = 300$$

The number of students in the science stream = 300

The number of students in the commerce stream = $472 - 300 = 172$

In School R

The total number of students = $(10/100) * 2000 = 200$

The number of students in the arts stream = $(12/100) * 400 = 48$

The sum of the number of students in the science and commerce stream together = $(200 - 48) = 152$

Let the number of students in the science stream be c .

So, the number of students in the commerce stream = $(152 - c)$

$$\text{ATD, } c - (152 - c) = 28$$

$$2c = 180$$

$$c = 90$$

The number of students in the science stream = 90

The number of students in the commerce stream = $152 - 90 = 62$

In School S

The total number of students = $(25/100) * 2000 = 500$

The number of students in the arts stream = $(18/100) * 400 = 72$

The sum of the number of students in the science and commerce stream together = $(500 - 72) = 428$

Let the number of students in the science stream be ' d '.

So, the number of students in the commerce stream = $(428 - d)$

$$\text{ATD, } d - (428 - d) = 292$$

$$2d = 720$$

$$d = 360$$

The number of students in the science stream = 360

The number of students in the commerce stream = $428 - 360 = 68$

In School T

The total number of students = $(15/100) * 2000 = 300$

The number of students in arts = $(14/100) * 400 = 56$

The sum of the number of students in science and commerce = $(300 - 56) = 244$

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Let the number of students in the science stream be 'e'.

So, the number of students in the commerce stream = $(244 - e)$

ATD, $e - (244 - e) = 60$

$2e = 304$

$e = 152$

The number of students in the science stream = 152

The number of students in the commerce stream = $244 - 152 = 92$

School	The total number of students	The number of students in the science stream	The number of students in the commerce stream	The number of students in the arts stream
P	400	228	76	96
Q	600	300	172	128
R	200	90	62	48
S	500	360	68	72
T	300	152	92	56
Total	2000	1130	470	400

Answer: B

In School Q,

The total number of students = 600

Let the number of students who passed and failed the examination be $4n$ and n respectively.

So, the number of students who did not appear in the examination = n

And, the number of students who appeared in the examination = $(600 - n)$

Now,

$4n + n = 600 - n$

$5n + n = 600$

$6n = 600$

$n = 100$

So, the number of students who did not appear in the examination = $n = 100$

Therefore, the required percentage = $\{(100/600) \times 100\} = 16.67\%$.

Hence, the correct answer is option B.

3. Questions

Let the total number of students in all five schools together be $500k$.

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So, the total number of students in the arts stream in all five schools together = $(1/5) * 500k = 100k$.

Now,

The total number of students in school S = $(25/100) * 500k = 125k$

The total number of students in the arts stream in school S = $(18/100) * 100k = 18k$

So, the sum of the number of students in the science and the commerce stream together in school S = $(125k - 18k) = 107k$

The number of students in the science stream = $(18k * 5) = 90k$

So, the number of students in the commerce stream = $(107k - 90k) = 17k$

Now,

$$90k - 17k = 292$$

$$73k = 292$$

$$k = 4$$

So, the total number of students in all five schools together = $500k = (500 * 4) = 2000$

And, the total number of students in the arts stream in all five schools together = $100k = (100 * 4) = 400$

In School P,

The total number of students = $(20/100) * 2000 = 400$

The number of students in the arts stream = $(24/100) * 400 = 96$

The sum of the number of students in the science and commerce stream together = $(400 - 96) = 304$

Let the number of students in the science stream be a.

So, the number of students in the commerce stream = $(304 - a)$

$$\text{ATD, } a - (304 - a) = 152$$

$$2a = 456$$

$$a = 228$$

The number of students in the science stream = 228

The number of students in the commerce stream = $304 - 228 = 76$

In School Q

The total number of students = $(30/100) * 2000 = 600$

The number of students in the arts stream = $(32/100) * 400 = 128$

The sum of the number of students in the science and commerce stream together = $(600 - 128) = 472$

Let the number of students in the science stream be b.

So, the number of students in the commerce stream = $(472 - b)$

$$\text{ATD, } b - (472 - b) = 128$$

$$2b = 600$$

$$b = 300$$

The number of students in the science stream = 300

The number of students in the commerce stream = $472 - 300 = 172$

In School R

The total number of students = $(10/100) * 2000 = 200$

The number of students in the arts stream = $(12/100) * 400 = 48$

The sum of the number of students in the science and commerce stream together = $(200 - 48) = 152$

Let the number of students in the science stream be c .

So, the number of students in the commerce stream = $(152 - c)$

$$\text{ATD, } c - (152 - c) = 28$$

$$2c = 180$$

$$c = 90$$

The number of students in the science stream = 90

The number of students in the commerce stream = $152 - 90 = 62$

In School S

The total number of students = $(25/100) * 2000 = 500$

The number of students in the arts stream = $(18/100) * 400 = 72$

The sum of the number of students in the science and commerce stream together = $(500 - 72) = 428$

Let the number of students in the science stream be ' d '.

So, the number of students in the commerce stream = $(428 - d)$

$$\text{ATD, } d - (428 - d) = 292$$

$$2d = 720$$

$$d = 360$$

The number of students in the science stream = 360

The number of students in the commerce stream = $428 - 360 = 68$

In School T

The total number of students = $(15/100) * 2000 = 300$

The number of students in arts = $(14/100) * 400 = 56$

The sum of the number of students in science and commerce = $(300 - 56) = 244$

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Let the number of students in the science stream be 'e'.

So, the number of students in the commerce stream = $(244 - e)$

ATD, $e - (244 - e) = 60$

$2e = 304$

$e = 152$

The number of students in the science stream = 152

The number of students in the commerce stream = $244 - 152 = 92$

School	The total number of students	The number of students in the science stream	The number of students in the commerce stream	The number of students in the arts stream
P	400	228	76	96
Q	600	300	172	128
R	200	90	62	48
S	500	360	68	72
T	300	152	92	56
Total	2000	1130	470	400

Answer: A

In School P (science)

The total number of students = 228

The number of male teachers = $(1/6) * 228 = 38$

So, the number of female teachers = $38 * (9/19) = 18$

The total number of teachers (males + females) = $(38+18) = 56$

$X = 56$

Therefore, $(X-15) = (56-15) = 41$.

Hence, the correct answer is option A.

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4. Questions

Let the total number of students in all five schools together be 500k.

So, the total number of students in the arts stream in all five schools together = $(1/5) * 500k = 100k$.

Now,

The total number of students in school S = $(25/100) * 500k = 125k$

The total number of students in the arts stream in school S = $(18/100) * 100k = 18k$

So, the sum of the number of students in the science and the commerce stream together in school S =

$$(125k - 18k) = 107k$$

$$\text{The number of students in the science stream} = (18k \times 5) = 90k$$

$$\text{So, the number of students in the commerce stream} = (107k - 90k) = 17k$$

Now,

$$90k - 17k = 292$$

$$73k = 292$$

$$k = 4$$

$$\text{So, the total number of students in all five schools together} = 500k = (500 \times 4) = 2000$$

$$\text{And, the total number of students in the arts stream in all five schools together} = 100k = (100 \times 4) = 400$$

In School P,

$$\text{The total number of students} = (20/100) \times 2000 = 400$$

$$\text{The number of students in the arts stream} = (24/100) \times 400 = 96$$

$$\text{The sum of the number of students in the science and commerce stream together} = (400 - 96) = 304$$

Let the number of students in the science stream be a .

$$\text{So, the number of students in the commerce stream} = (304 - a)$$

$$\text{ATD, } a - (304 - a) = 152$$

$$2a = 456$$

$$a = 228$$

$$\text{The number of students in the science stream} = 228$$

$$\text{The number of students in the commerce stream} = 304 - 228 = 76$$

In School Q

$$\text{The total number of students} = (30/100) \times 2000 = 600$$

$$\text{The number of students in the arts stream} = (32/100) \times 400 = 128$$

$$\text{The sum of the number of students in the science and commerce stream together} = (600 - 128) = 472$$

Let the number of students in the science stream be b .

$$\text{So, the number of students in the commerce stream} = (472 - b)$$

$$\text{ATD, } b - (472 - b) = 128$$

$$2b = 600$$

$$b = 300$$

$$\text{The number of students in the science stream} = 300$$

$$\text{The number of students in the commerce stream} = 472 - 300 = 172$$

In School R

The total number of students = $(10/100) * 2000 = 200$

The number of students in the arts stream = $(12/100) * 400 = 48$

The sum of the number of students in the science and commerce stream together = $(200 - 48) = 152$

Let the number of students in the science stream be c .

So, the number of students in the commerce stream = $(152 - c)$

ATD, $c - (152 - c) = 28$

$2c = 180$

$c = 90$

The number of students in the science stream = 90

The number of students in the commerce stream = $152 - 90 = 62$

In School S

The total number of students = $(25/100) * 2000 = 500$

The number of students in the arts stream = $(18/100) * 400 = 72$

The sum of the number of students in the science and commerce stream together = $(500 - 72) = 428$

Let the number of students in the science stream be ' d '.

So, the number of students in the commerce stream = $(428 - d)$

ATD, $d - (428 - d) = 292$

$2d = 720$

$d = 360$

The number of students in the science stream = 360

The number of students in the commerce stream = $428 - 360 = 68$

In School T

The total number of students = $(15/100) * 2000 = 300$

The number of students in arts = $(14/100) * 400 = 56$

The sum of the number of students in science and commerce = $(300 - 56) = 244$

Let the number of students in the science stream be ' e '.

So, the number of students in the commerce stream = $(244 - e)$

ATD, $e - (244 - e) = 60$

$2e = 304$

$e = 152$

@QuizByMickey

The number of students in the science stream = 152

The number of students in the commerce stream = $244 - 152 = 92$

School	The total number of students	The number of students in the science stream	The number of students in the commerce stream	The number of students in the arts stream
P	400	228	76	96
Q	600	300	172	128
R	200	90	62	48
S	500	360	68	72
T	300	152	92	56
Total	2000	1130	470	400

Answer: D

In School U,

The total number of students = $(1/3) * 300 = 100$

The number of students in the arts stream = 22

Therefore, the number of students in the commerce stream in school U = $(5/13) * (100 - 22) = (5/13) * 78 = 30$.

Hence, the correct answer is option D.

5. Questions

Let the total number of students in all five schools together be 500k.

So, the total number of students in the arts stream in all five schools together = $(1/5) * 500k = 100k$.

Now,

The total number of students in school S = $(25/100) * 500k = 125k$

The total number of students in the arts stream in school S = $(18/100) * 100k = 18k$

So, the sum of the number of students in the science and the commerce stream together in school S = $(125k - 18k) = 107k$

The number of students in the science stream = $(18k * 5) = 90k$

So, the number of students in the commerce stream = $(107k - 90k) = 17k$

Now,

$90k - 17k = 292$

$73k = 292$

$k = 4$

So, the total number of students in all five schools together = $500k = (500 \times 4) = 2000$

And, the total number of students in the arts stream in all five schools together = $100k = (100 \times 4) = 400$

In School P,

The total number of students = $(20/100) \times 2000 = 400$

The number of students in the arts stream = $(24/100) \times 400 = 96$

The sum of the number of students in the science and commerce stream together = $(400 - 96) = 304$

Let the number of students in the science stream be a.

So, the number of students in the commerce stream = $(304 - a)$

ATD, $a - (304 - a) = 152$

$2a = 456$

$a = 228$

The number of students in the science stream = 228

The number of students in the commerce stream = $304 - 228 = 76$

In School Q

The total number of students = $(30/100) \times 2000 = 600$

The number of students in the arts stream = $(32/100) \times 400 = 128$

The sum of the number of students in the science and commerce stream together = $(600 - 128) = 472$

Let the number of students in the science stream be b.

So, the number of students in the commerce stream = $(472 - b)$

ATD, $b - (472 - b) = 128$

$2b = 600$

$b = 300$

The number of students in the science stream = 300

The number of students in the commerce stream = $472 - 300 = 172$

In School R

The total number of students = $(10/100) \times 2000 = 200$

The number of students in the arts stream = $(12/100) \times 400 = 48$

The sum of the number of students in the science and commerce stream together = $(200 - 48) = 152$

Let the number of students in the science stream be c.

So, the number of students in the commerce stream = $(152 - c)$

ATD, $c - (152 - c) = 28$

@QuizByMickey

$$2c = 180$$

$$c = 90$$

The number of students in the science stream = 90

The number of students in the commerce stream = $152 - 90 = 62$

In School S

The total number of students = $(25/100) * 2000 = 500$

The number of students in the arts stream = $(18/100) * 400 = 72$

The sum of the number of students in the science and commerce stream together = $(500 - 72) = 428$

Let the number of students in the science stream be 'd'.

So, the number of students in the commerce stream = $(428 - d)$

$$\text{ATD, } d - (428 - d) = 292$$

$$2d = 720$$

$$d = 360$$

The number of students in the science stream = 360

The number of students in the commerce stream = $428 - 360 = 68$

In School T

The total number of students = $(15/100) * 2000 = 300$

The number of students in arts = $(14/100) * 400 = 56$

The sum of the number of students in science and commerce = $(300 - 56) = 244$

Let the number of students in the science stream be 'e'.

So, the number of students in the commerce stream = $(244 - e)$

$$\text{ATD, } e - (244 - e) = 60$$

$$2e = 304$$

$$e = 152$$

The number of students in the science stream = 152

The number of students in the commerce stream = $244 - 152 = 92$

@QuizByMickey

School	The total number of students	The number of students in the science stream	The number of students in the commerce stream	The number of students in the arts stream
P	400	228	76	96
Q	600	300	172	128
R	200	90	62	48
S	500	360	68	72
T	300	152	92	56
Total	2000	1130	470	400

Answer: B

I:

The number of students in the commerce stream in school P = 72

The number of students in the arts stream in school P = 96

According to the statement,

$$96 - 72 = 24$$

The number of students in the commerce stream in school S = 68

The number of students in the arts stream in school S = 72

According to the statement,

$$72 - 68 = 4$$

So, I is definitely true.

II:

The number of students in the science stream in school Q = 300

The total number of students in school Q = 600

$$(300/600) = 1/2$$

So, II is definitely not true.

III:

The number of students in the commerce stream in school R = 62

The number of students in the commerce stream in school T = 92

$$\text{The required difference} = (92 - 62) = 30$$

So, III is definitely true.

Therefore, both I and III are definitely true.

Hence, the correct answer is option B.

@QuizByMickey

6. Questions

Let the total number of students in all five schools together be 500k.

So, the total number of students in the arts stream in all five schools together = $(1/5) * 500k = 100k$.

Now,

The total number of students in school S = $(25/100) * 500k = 125k$

The total number of students in the arts stream in school S = $(18/100) * 100k = 18k$

So, the sum of the number of students in the science and the commerce stream together in school S = $(125k - 18k) = 107k$

The number of students in the science stream = $(18k * 5) = 90k$

So, the number of students in the commerce stream = $(107k - 90k) = 17k$

Now,

$$90k - 17k = 292$$

$$73k = 292$$

$$k = 4$$

So, the total number of students in all five schools together = $500k = (500 * 4) = 2000$

And, the total number of students in the arts stream in all five schools together = $100k = (100 * 4) = 400$

In School P,

The total number of students = $(20/100) * 2000 = 400$

The number of students in the arts stream = $(24/100) * 400 = 96$

The sum of the number of students in the science and commerce stream together = $(400 - 96) = 304$

Let the number of students in the science stream be a.

So, the number of students in the commerce stream = $(304 - a)$

$$\text{ATD, } a - (304 - a) = 152$$

$$2a = 456$$

$$a = 228$$

The number of students in the science stream = 228

The number of students in the commerce stream = $304 - 228 = 76$

In School Q

The total number of students = $(30/100) * 2000 = 600$

The number of students in the arts stream = $(32/100) * 600 = 128$

The sum of the number of students in the science and commerce stream together = $(600 - 128) = 472$

@QuizByMickey

Let the number of students in the science stream be b .

So, the number of students in the commerce stream = $(472 - b)$

ATD, $b - (472 - b) = 128$

$$2b = 600$$

$$b = 300$$

The number of students in the science stream = 300

The number of students in the commerce stream = $472 - 300 = 172$

In School R

The total number of students = $(10/100) * 2000 = 200$

The number of students in the arts stream = $(12/100) * 400 = 48$

The sum of the number of students in the science and commerce stream together = $(200 - 48) = 152$

Let the number of students in the science stream be c .

So, the number of students in the commerce stream = $(152 - c)$

ATD, $c - (152 - c) = 28$

$$2c = 180$$

$$c = 90$$

The number of students in the science stream = 90

The number of students in the commerce stream = $152 - 90 = 62$

In School S

The total number of students = $(25/100) * 2000 = 500$

The number of students in the arts stream = $(18/100) * 400 = 72$

The sum of the number of students in the science and commerce stream together = $(500 - 72) = 428$

Let the number of students in the science stream be ' d '.

So, the number of students in the commerce stream = $(428 - d)$

ATD, $d - (428 - d) = 292$

$$2d = 720$$

$$d = 360$$

The number of students in the science stream = 360

The number of students in the commerce stream = $428 - 360 = 68$

In School T

The total number of students = $(15/100) * 2000 = 300$

@QuizByMickey

The number of students in arts = $(14/100) * 400 = 56$

The sum of the number of students in science and commerce = $(300-56) = 244$

Let the number of students in the science stream be 'e'.

So, the number of students in the commerce stream = $(244 - e)$

ATD, $e - (244 - e) = 60$

$2e = 304$

$e = 152$

The number of students in the science stream = 152

The number of students in the commerce stream = $244 - 152 = 92$

School	The total number of students	The number of students in the science stream	The number of students in the commerce stream	The number of students in the arts stream
P	400	228	76	96
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R	200	90	62	48
S	500	360	68	72
T	300	152	92	56
Total	2000	1130	470	400

Answer: A

In School R,

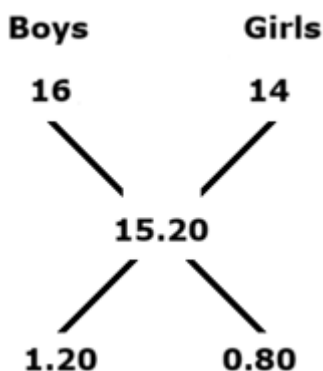
The total number of students = 200

So, the average age of all students = $(3040/200) = 15.20$ years

The average age of all boys = 16 years

The average age of all girls = 14 years

Now, using the allegation method, we get,



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So, the ratio of the number of boys and girls = $1.20:0.80 = 3:2$.

Therefore, the number of girls = $(2/5) * 200 = 80$.

Another method:

In school R,

Let the number of girls be 'z'

So, the number of boys = $200 - z$

According to the question,

$$(200 - z) * 16 + (z * 14) = 3040$$

$$3200 - 16z + 14z = 3040$$

$$160 = 2z$$

$$z = 80$$

The number of girls = 80

Hence, the correct answer is option A.

7. Questions

In School A,

Let the total number of people be P.

The number of illiterate people = 300

The number of literate people = $(P-300)$

The number of graduates = 500

The number of post-graduates = $\{(P-300)-500\} = (P-800)$

Now,

$$P/(P-800) = 3/1$$

$$P = 3P - 2400$$

$$2P = 2400$$

$$P = 1200$$

Therefore,

The total number of people = $P = 1200$

The number of literate people = $(P-300) = (1200-300) = 900$

The number of post-graduates = $(P-800) = (1200-800) = 400$

In School B,

Let the total number of people be Q.

@QuizByMickey

The number of illiterate people = 400

The number of literate people = (Q-400)

The number of graduates = 1000

The number of post-graduates = $\{(Q-400)-1000\} = (Q-1400)$

Now,

$$Q/(Q-1400) = 10/3$$

$$3Q = 10Q - 14000$$

$$7Q = 14000$$

$$Q = 2000$$

Therefore,

The total number of people = $Q = 2000$

The number of literate people = $(Q-400) = (2000-400) = 1600$

The number of post-graduates = $(Q-1400) = (2000-1400) = 600$

In School C,

Let the total number of people be R.

The number of illiterate people = 800

The number of literate people = $(R-800)$

The number of graduates = 1600

The number of post-graduates = $\{(R-800)-1600\} = (R-2400)$

Now,

$$R/(R-2400) = 4/1$$

$$R = 4R - 9600$$

$$3R = 9600$$

$$R = 3200$$

Therefore,

The total number of people = $R = 3200$

The number of literate people = $(R-800) = (3200-800) = 2400$

The number of graduates = 1600

The number of post-graduates = $(R-2400) = (3200-2400) = 800$

In School D,

Let the total number of people be S.

@QuizByMickey

The number of illiterate people = 400

The number of literate people = (S-400)

The number of graduates = 300

The number of post-graduates = $\{(S-400)-300\} = (S-700)$

Now,

$$S/(S-700) = 10/3$$

$$3S = 10S - 7000$$

$$7S = 7000$$

$$S = 1000$$

Therefore,

The total number of people = $S = 1000$

The number of literate people = $(S-400) = (1000-400) = 600$

The number of post-graduates = $(S-700) = (1000-700) = 300$

In School E,

Let the total number of people be T.

The number of illiterate people = 1000

The number of literate people = $(T-1000)$

The number of graduates = 1600

The number of post-graduates = $\{(T-1000)-1600\} = (T-2600)$

Now,

$$T/(T-2600) = 15/2$$

$$2T = 15T - 39000$$

$$13T = 39000$$

$$T = 3000$$

Therefore,

The total number of people = $T = 3000$

The number of literate people = $(T-1000) = (3000-1000) = 2000$

The number of post-graduates = $(T-2600) = (3000-2600) = 400$

@QuizByMickey

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduates	The number of post-graduates
A	1200	900	300	500	400
B	2000	1600	400	1000	600
C	3200	2400	800	1600	800
D	1000	600	400	300	300
E	3000	2000	1000	1600	400
Total	10400	7500	2900	5000	2500

Answer: C

In Village A,

The number of literate people = 900

The number of literate males = $(2/3) \times 900 = 600$

The number of literate females = $(900-600) = 300$

The number of post-graduates = 400

The number of female post-graduates = $(3/8) \times 400 = 150$

So, the number of female graduates = $(300-150) = 150$

And, the number of graduates = 500

Therefore, the required percentage = $\{(150/500) \times 100\} = 30\%$.

Hence, the correct answer is option C.

8. Questions

In School A,

Let the total number of people be P.

The number of illiterate people = 300

The number of literate people = $(P-300)$

The number of graduates = 500

The number of post-graduates = $\{(P-300)-500\} = (P-800)$

Now,

$$P/(P-800) = 3/1$$

$$P = 3P - 2400$$

$$2P = 2400$$

$$P = 1200$$

@QuizByMickey

Therefore,

The total number of people = $P = 1200$

The number of literate people = $(P-300) = (1200-300) = 900$

The number of post-graduates = $(P-800) = (1200-800) = 400$

In School B,

Let the total number of people be Q .

The number of illiterate people = 400

The number of literate people = $(Q-400)$

The number of graduates = 1000

The number of post-graduates = $\{(Q-400)-1000\} = (Q-1400)$

Now,

$$Q/(Q-1400) = 10/3$$

$$3Q = 10Q - 14000$$

$$7Q = 14000$$

$$Q = 2000$$

Therefore,

The total number of people = $Q = 2000$

The number of literate people = $(Q-400) = (2000-400) = 1600$

The number of post-graduates = $(Q-1400) = (2000-1400) = 600$

In School C,

Let the total number of people be R .

The number of illiterate people = 800

The number of literate people = $(R-800)$

The number of graduates = 1600

The number of post-graduates = $\{(R-800)-1600\} = (R-2400)$

Now,

$$R/(R-2400) = 4/1$$

$$R = 4R - 9600$$

$$3R = 9600$$

$$R = 3200$$

Therefore,

@QuizByMickey

The total number of people = $R = 3200$

The number of literate people = $(R-800) = (3200-800) = 2400$

The number of graduates = 1600

The number of post-graduates = $(R-2400) = (3200-2400) = 800$

In School D,

Let the total number of people be S .

The number of illiterate people = 400

The number of literate people = $(S-400)$

The number of graduates = 300

The number of post-graduates = $\{(S-400)-300\} = (S-700)$

Now,

$$S/(S-700) = 10/3$$

$$3S = 10S - 7000$$

$$7S = 7000$$

$$S = 1000$$

Therefore,

The total number of people = $S = 1000$

The number of literate people = $(S-400) = (1000-400) = 600$

The number of post-graduates = $(S-700) = (1000-700) = 300$

In School E,

Let the total number of people be T .

The number of illiterate people = 1000

The number of literate people = $(T-1000)$

The number of graduates = 1600

The number of post-graduates = $\{(T-1000)-1600\} = (T-2600)$

Now,

$$T/(T-2600) = 15/2$$

$$2T = 15T - 39000$$

$$13T = 39000$$

$$T = 3000$$

Therefore,

@QuizByMickey

The total number of people = $T = 3000$

The number of literate people = $(T-1000) = (3000-1000) = 2000$

The number of post-graduates = $(T-2600) = (3000-2600) = 400$

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduates	The number of post-graduates
A	1200	900	300	500	400
B	2000	1600	400	1000	600
C	3200	2400	800	1600	800
D	1000	600	400	300	300
E	3000	2000	1000	1600	400
Total	10400	7500	2900	5000	2500

Answer: A

In Village B,

The number of post-graduates = 600

The number of unemployed post-graduates = $(1/3) * 600 = 200$ (**Note: $33.33\% = 1/3$**)

The number of employed post-graduates = $(600-200) = 400$

The number of graduates = 1000

The number of unemployed graduates = $(200-80) = 120$

The number of employed graduates = $(1000-120) = 880$

Therefore, the required ratio = $880:400 = 11:5$.

Hence, the correct answer is option A.

9. Questions

In School A,

Let the total number of people be P.

The number of illiterate people = 300

The number of literate people = $(P-300)$

The number of graduates = 500

The number of post-graduates = $\{(P-300)-500\} = (P-800)$

Now,

$$P/(P-800) = 3/1$$

$$P = 3P - 2400$$

@QuizByMickey

$$2P = 2400$$

$$P = 1200$$

Therefore,

The total number of people = $P = 1200$

The number of literate people = $(P-300) = (1200-300) = 900$

The number of post-graduates = $(P-800) = (1200-800) = 400$

In School B,

Let the total number of people be Q .

The number of illiterate people = 400

The number of literate people = $(Q-400)$

The number of graduates = 1000

The number of post-graduates = $\{(Q-400)-1000\} = (Q-1400)$

Now,

$$Q/(Q-1400) = 10/3$$

$$3Q = 10Q - 14000$$

$$7Q = 14000$$

$$Q = 2000$$

Therefore,

The total number of people = $Q = 2000$

The number of literate people = $(Q-400) = (2000-400) = 1600$

The number of post-graduates = $(Q-1400) = (2000-1400) = 600$

In School C,

Let the total number of people be R .

The number of illiterate people = 800

The number of literate people = $(R-800)$

The number of graduates = 1600

The number of post-graduates = $\{(R-800)-1600\} = (R-2400)$

Now,

$$R/(R-2400) = 4/1$$

$$R = 4R - 9600$$

$$3R = 9600$$

@QuizByMickey

$$R = 3200$$

Therefore,

$$\text{The total number of people} = R = 3200$$

$$\text{The number of literate people} = (R-800) = (3200-800) = 2400$$

$$\text{The number of graduates} = 1600$$

$$\text{The number of post-graduates} = (R-2400) = (3200-2400) = 800$$

In School D,

Let the total number of people be S.

$$\text{The number of illiterate people} = 400$$

$$\text{The number of literate people} = (S-400)$$

$$\text{The number of graduates} = 300$$

$$\text{The number of post-graduates} = \{(S-400)-300\} = (S-700)$$

Now,

$$S/(S-700) = 10/3$$

$$3S = 10S - 7000$$

$$7S = 7000$$

$$S = 1000$$

Therefore,

$$\text{The total number of people} = S = 1000$$

$$\text{The number of literate people} = (S-400) = (1000-400) = 600$$

$$\text{The number of post-graduates} = (S-700) = (1000-700) = 300$$

In School E,

Let the total number of people be T.

$$\text{The number of illiterate people} = 1000$$

$$\text{The number of literate people} = (T-1000)$$

$$\text{The number of graduates} = 1600$$

$$\text{The number of post-graduates} = \{(T-1000)-1600\} = (T-2600)$$

Now,

$$T/(T-2600) = 15/2$$

$$2T = 15T - 39000$$

$$13T = 39000$$

@QuizByMickey

$$T = 3000$$

Therefore,

$$\text{The total number of people} = T = 3000$$

$$\text{The number of literate people} = (T - 1000) = (3000 - 1000) = 2000$$

$$\text{The number of post-graduates} = (T - 2600) = (3000 - 2600) = 400$$

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduates	The number of post-graduates
A	1200	900	300	500	400
B	2000	1600	400	1000	600
C	3200	2400	800	1600	800
D	1000	600	400	300	300
E	3000	2000	1000	1600	400
Total	10400	7500	2900	5000	2500

Answer: B

In Village F,

$$\text{The number of literate people} = (2400 + 600) = 3000$$

$$\text{The number of graduates} = (3/5) * 3000 = 1800$$

$$\text{The number of post-graduates} = (2/5) * 3000 = 1200$$

$$\text{So, the number of illiterate people} = 1200 * (100/120) = 1000$$

$$\text{Therefore, the total number of people in village F} = (3000 + 1000) = 4000.$$

Hence, the correct answer is option B.

10. Questions

In School A,

Let the total number of people be P.

$$\text{The number of illiterate people} = 300$$

$$\text{The number of literate people} = (P - 300)$$

$$\text{The number of graduates} = 500$$

$$\text{The number of post-graduates} = \{(P - 300) - 500\} = (P - 800)$$

Now,

$$P / (P - 800) = 3/1$$

$$P = 3P - 2400$$

@QuizByMickey

$$2P = 2400$$

$$P = 1200$$

Therefore,

$$\text{The total number of people} = P = 1200$$

$$\text{The number of literate people} = (P-300) = (1200-300) = 900$$

$$\text{The number of post-graduates} = (P-800) = (1200-800) = 400$$

In School B,

Let the total number of people be Q.

$$\text{The number of illiterate people} = 400$$

$$\text{The number of literate people} = (Q-400)$$

$$\text{The number of graduates} = 1000$$

$$\text{The number of post-graduates} = \{(Q-400)-1000\} = (Q-1400)$$

Now,

$$Q/(Q-1400) = 10/3$$

$$3Q = 10Q - 14000$$

$$7Q = 14000$$

$$Q = 2000$$

Therefore,

$$\text{The total number of people} = Q = 2000$$

$$\text{The number of literate people} = (Q-400) = (2000-400) = 1600$$

$$\text{The number of post-graduates} = (Q-1400) = (2000-1400) = 600$$

In School C,

Let the total number of people be R.

$$\text{The number of illiterate people} = 800$$

$$\text{The number of literate people} = (R-800)$$

$$\text{The number of graduates} = 1600$$

$$\text{The number of post-graduates} = \{(R-800)-1600\} = (R-2400)$$

Now,

$$R/(R-2400) = 4/1$$

$$R = 4R - 9600$$

$$3R = 9600$$

@QuizByMickey

$$R = 3200$$

Therefore,

$$\text{The total number of people} = R = 3200$$

$$\text{The number of literate people} = (R-800) = (3200-800) = 2400$$

$$\text{The number of graduates} = 1600$$

$$\text{The number of post-graduates} = (R-2400) = (3200-2400) = 800$$

In School D,

Let the total number of people be S.

$$\text{The number of illiterate people} = 400$$

$$\text{The number of literate people} = (S-400)$$

$$\text{The number of graduates} = 300$$

$$\text{The number of post-graduates} = \{(S-400)-300\} = (S-700)$$

Now,

$$S/(S-700) = 10/3$$

$$3S = 10S - 7000$$

$$7S = 7000$$

$$S = 1000$$

Therefore,

$$\text{The total number of people} = S = 1000$$

$$\text{The number of literate people} = (S-400) = (1000-400) = 600$$

$$\text{The number of post-graduates} = (S-700) = (1000-700) = 300$$

In School E,

Let the total number of people be T.

$$\text{The number of illiterate people} = 1000$$

$$\text{The number of literate people} = (T-1000)$$

$$\text{The number of graduates} = 1600$$

$$\text{The number of post-graduates} = \{(T-1000)-1600\} = (T-2600)$$

Now,

$$T/(T-2600) = 15/2$$

$$2T = 15T - 39000$$

$$13T = 39000$$

@QuizByMickey

$$T = 3000$$

Therefore,

$$\text{The total number of people} = T = 3000$$

$$\text{The number of literate people} = (T - 1000) = (3000 - 1000) = 2000$$

$$\text{The number of post-graduates} = (T - 2600) = (3000 - 2600) = 400$$

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduates	The number of post-graduates
A	1200	900	300	500	400
B	2000	1600	400	1000	600
C	3200	2400	800	1600	800
D	1000	600	400	300	300
E	3000	2000	1000	1600	400
Total	10400	7500	2900	5000	2500

Answer: E

In Village D,

$$\text{The total number of people} = 1000$$

$$\text{The number of females} = (30/100) * 1000 = 300$$

$$\text{The number of males} = (1000 - 300) = 700$$

$$\text{The number of people who are literate} = 600$$

$$\text{The number of males who are literate} = (9/10) * 600 = 540$$

$$\text{Therefore, the number of males who are illiterate} = (700 - 540) = 160.$$

Hence, the correct answer is option E.

11. Questions

In School A,

Let the total number of people be P.

$$\text{The number of illiterate people} = 300$$

$$\text{The number of literate people} = (P - 300)$$

$$\text{The number of graduates} = 500$$

$$\text{The number of post-graduates} = \{(P - 300) - 500\} = (P - 800)$$

Now,

$$P/(P - 800) = 3/1$$

@QuizByMickey

$$P = 3P - 2400$$

$$2P = 2400$$

$$P = 1200$$

Therefore,

The total number of people = $P = 1200$

The number of literate people = $(P-300) = (1200-300) = 900$

The number of post-graduates = $(P-800) = (1200-800) = 400$

In School B,

Let the total number of people be Q .

The number of illiterate people = 400

The number of literate people = $(Q-400)$

The number of graduates = 1000

The number of post-graduates = $\{(Q-400)-1000\} = (Q-1400)$

Now,

$$Q/(Q-1400) = 10/3$$

$$3Q = 10Q - 14000$$

$$7Q = 14000$$

$$Q = 2000$$

Therefore,

The total number of people = $Q = 2000$

The number of literate people = $(Q-400) = (2000-400) = 1600$

The number of post-graduates = $(Q-1400) = (2000-1400) = 600$

In School C,

Let the total number of people be R .

The number of illiterate people = 800

The number of literate people = $(R-800)$

The number of graduates = 1600

The number of post-graduates = $\{(R-800)-1600\} = (R-2400)$

Now,

$$R/(R-2400) = 4/1$$

$$R = 4R - 9600$$

@QuizByMickey

$$3R = 9600$$

$$R = 3200$$

Therefore,

$$\text{The total number of people} = R = 3200$$

$$\text{The number of literate people} = (R-800) = (3200-800) = 2400$$

$$\text{The number of graduates} = 1600$$

$$\text{The number of post-graduates} = (R-2400) = (3200-2400) = 800$$

In School D,

Let the total number of people be S.

$$\text{The number of illiterate people} = 400$$

$$\text{The number of literate people} = (S-400)$$

$$\text{The number of graduates} = 300$$

$$\text{The number of post-graduates} = \{(S-400)-300\} = (S-700)$$

Now,

$$S/(S-700) = 10/3$$

$$3S = 10S - 7000$$

$$7S = 7000$$

$$S = 1000$$

Therefore,

$$\text{The total number of people} = S = 1000$$

$$\text{The number of literate people} = (S-400) = (1000-400) = 600$$

$$\text{The number of post-graduates} = (S-700) = (1000-700) = 300$$

In School E,

Let the total number of people be T.

$$\text{The number of illiterate people} = 1000$$

$$\text{The number of literate people} = (T-1000)$$

$$\text{The number of graduates} = 1600$$

$$\text{The number of post-graduates} = \{(T-1000)-1600\} = (T-2600)$$

Now,

$$T/(T-2600) = 15/2$$

$$2T = 15T - 39000$$

@QuizByMickey

$$13T = 39000$$

$$T = 3000$$

Therefore,

$$\text{The total number of people} = T = 3000$$

$$\text{The number of literate people} = (T - 1000) = (3000 - 1000) = 2000$$

$$\text{The number of post-graduates} = (T - 2600) = (3000 - 2600) = 400$$

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduates	The number of post-graduates
A	1200	900	300	500	400
B	2000	1600	400	1000	600
C	3200	2400	800	1600	800
D	1000	600	400	300	300
E	3000	2000	1000	1600	400
Total	10400	7500	2900	5000	2500

Answer: B

X:

$$\text{The number of literate people in village A} = 900$$

$$\text{The number of literate people in village D} = 600$$

$$\text{So, } X = \{(900 + 600) / 2\} = (1500 / 2) = 750$$

Y:

$$\text{The total number of people in village B} = 2000$$

$$\text{The total number of people in village C} = 3200$$

$$\text{So, } Y = (3200 - 2000) = 1200$$

Z:

$$\text{The number of post-graduates in village A} = 400$$

$$\text{The number of post-graduates in village E} = 400$$

$$\text{So, } Z = (400 + 400) = 800$$

$$\text{Therefore, } X < Y > Z.$$

Hence, the correct answer is option B.

12. Questions

In School A,

@QuizByMickey

Let the total number of people be P.

The number of illiterate people = 300

The number of literate people = (P-300)

The number of graduates = 500

The number of post-graduates = $\{(P-300)-500\} = (P-800)$

Now,

$$P/(P-800) = 3/1$$

$$P = 3P - 2400$$

$$2P = 2400$$

$$P = 1200$$

Therefore,

The total number of people = P = 1200

The number of literate people = (P-300) = (1200-300) = 900

The number of post-graduates = (P-800) = (1200-800) = 400

In School B,

Let the total number of people be Q.

The number of illiterate people = 400

The number of literate people = (Q-400)

The number of graduates = 1000

The number of post-graduates = $\{(Q-400)-1000\} = (Q-1400)$

Now,

$$Q/(Q-1400) = 10/3$$

$$3Q = 10Q - 14000$$

$$7Q = 14000$$

$$Q = 2000$$

Therefore,

The total number of people = Q = 2000

The number of literate people = (Q-400) = (2000-400) = 1600

The number of post-graduates = (Q-1400) = (2000-1400) = 600

In School C,

Let the total number of people be R.

@QuizByMickey

The number of illiterate people = 800

The number of literate people = (R-800)

The number of graduates = 1600

The number of post-graduates = $\{(R-800)-1600\} = (R-2400)$

Now,

$$R/(R-2400) = 4/1$$

$$R = 4R - 9600$$

$$3R = 9600$$

$$R = 3200$$

Therefore,

The total number of people = $R = 3200$

The number of literate people = $(R-800) = (3200-800) = 2400$

The number of graduates = 1600

The number of post-graduates = $(R-2400) = (3200-2400) = 800$

In School D,

Let the total number of people be S.

The number of illiterate people = 400

The number of literate people = $(S-400)$

The number of graduates = 300

The number of post-graduates = $\{(S-400)-300\} = (S-700)$

Now,

$$S/(S-700) = 10/3$$

$$3S = 10S - 7000$$

$$7S = 7000$$

$$S = 1000$$

Therefore,

The total number of people = $S = 1000$

The number of literate people = $(S-400) = (1000-400) = 600$

The number of post-graduates = $(S-700) = (1000-700) = 300$

In School E,

Let the total number of people be T.

@QuizByMickey

The number of illiterate people = 1000

The number of literate people = (T-1000)

The number of graduates = 1600

The number of post-graduates = $\{(T-1000)-1600\} = (T-2600)$

Now,

$$T/(T-2600) = 15/2$$

$$2T = 15T - 39000$$

$$13T = 39000$$

$$T = 3000$$

Therefore,

The total number of people = T = 3000

The number of literate people = (T-1000) = (3000-1000) = 2000

The number of post-graduates = (T-2600) = (3000-2600) = 400

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduates	The number of post-graduates
A	1200	900	300	500	400
B	2000	1600	400	1000	600
C	3200	2400	800	1600	800
D	1000	600	400	300	300
E	3000	2000	1000	1600	400
Total	10400	7500	2900	5000	2500

Answer: C

In Village C,

The total number of post-graduates = 800

The number of post-graduates in private college = N

The number of post-graduates in government college = (N+M)

Now,

$$N + (N+M) = 800$$

$$2N + M = 800 \text{ -----}>(1)$$

And,

$$N - M = 100 \text{ -----}>(2)$$

@QuizByMickey

By solving equations (1) and (2), we get,

$$2N + M = 800$$

$$2N - 2M = 200$$

$$3M = 600$$

$$M = 200$$

Now, put the value of M in equation (1), we get,

$$2N + M = 800$$

$$2N + 200 = 800$$

$$2N = 600$$

$$N = 300$$

Therefore, $(2M - N) = (2 \times 200) - 300 = (400 - 300) = 100$.

Hence, the correct answer is option C.

13. Questions

The time taken to complete the whole work by,

$$(P+Q) = 12 \text{ days}$$

$$(R+S+T) = (150/31) \text{ days}$$

$$(P+S) = \{(10/3) \times 2\} = (20/3) \text{ days}$$

$$T = (30/2) = 15 \text{ days}$$

$$(P+R) = (100/9) \text{ days}$$

The total work = 300 units
(L.C.M. of the numerator
of the time taken)

So, the efficiency of,

$$(P+Q) = (300/12) = 25 \text{ units}$$

$$(R+S+T) = \{300/(150/31)\} = 62 \text{ units}$$

$$(P+S) = \{300/(20/3)\} = 45 \text{ units}$$

$$T = (300/15) = 20 \text{ units}$$

$$(P+R) = \{300/(100/9)\} = 27 \text{ units}$$

So,

$$(R+S) = (62-20) = 42 \text{ units}$$

$$S = 42 - R$$

$$P + 42 - R = 45 \text{ units}$$

$$P - R = 3 \text{ -----} > (1)$$

$$P + R = 27 \text{ ----} > (2)$$

@QuizByMickey

By solving equations (1) and (2), we get,

$$P = 30/2 = 15$$

$$R = 27 - 15$$

$$R = 12 \text{ units}$$

$$P+Q = 25$$

$$15 + Q = 25$$

$$Q = 10$$

Also,

$$(R+S+T) = 62$$

$$12 + S + 20 = 62$$

$$S = 62 - 32$$

$$S = 30 \text{ units}$$

Now, the individual efficiency of,

$$P = 15 \text{ units}$$

$$Q = 10 \text{ units}$$

$$R = 12 \text{ units}$$

$$S = 30 \text{ units}$$

$$T = 20 \text{ units}$$

So, the time taken to complete the whole work alone by,

$$P = (300/15) = 20 \text{ days}$$

$$Q = (300/10) = 30 \text{ days}$$

$$R = (300/12) = 25 \text{ days}$$

$$S = (300/30) = 10 \text{ days}$$

$$T = (300/20) = 15 \text{ days}$$

Answer: A

Now, according to the question,

The reduced efficiency of $P = \{15 - (X\% \text{ of } 15)\} = (15 - 0.15X)$ units

The increased efficiency of $Q = [10 + \{(2X+10)\% \text{ of } 10\}] = \{10 + (0.20X+1)\} = (11+0.20X)$

Now,

$$[(15-0.15X) + (11+0.20X)] * (100/9) = 300$$

$$(26+0.05X) = 300 * (9/100)$$

@QuizByMickey

$$26 + 0.05X = 27$$

$$0.05X = 1$$

$$X = 20$$

Therefore, the value of $(X/5) = (20/5) = 4$.

Hence, the correct answer is option A.

14. Questions

The time taken to complete the whole work by,

$$(P+Q) = 12 \text{ days}$$

$$(R+S+T) = (150/31) \text{ days}$$

$$(P+S) = \{(10/3) \times 2\} = (20/3) \text{ days}$$

$$T = (30/2) = 15 \text{ days}$$

$$(P+R) = (100/9) \text{ days}$$

The total work = 300 units
 (L.C.M. of the numerator
 of the time taken)

So, the efficiency of,

$$(P+Q) = (300/12) = 25 \text{ units}$$

$$(R+S+T) = \{300/(150/31)\} = 62 \text{ units}$$

$$(P+S) = \{300/(20/3)\} = 45 \text{ units}$$

$$T = (300/15) = 20 \text{ units}$$

$$(P+R) = \{300/(100/9)\} = 27 \text{ units}$$

So,

$$(R+S) = (62-20) = 42 \text{ units}$$

$$S = 42 - R$$

$$P + 42 - R = 45 \text{ units}$$

$$P - R = 3 \text{ -----} > (1)$$

$$P + R = 27 \text{ ----} > (2)$$

By solving equations (1) and (2), we get,

$$P = 30/2 = 15$$

$$R = 27 - 15$$

$$R = 12 \text{ units}$$

$$P+Q = 25$$

$$15 + Q = 25$$

$$Q = 10$$

@QuizByMickey

Also,

$$(R+S+T) = 62$$

$$12 + S + 20 = 62$$

$$S = 62 - 32$$

$$S = 30 \text{ units}$$

Now, the individual efficiency of,

$$P = 15 \text{ units}$$

$$Q = 10 \text{ units}$$

$$R = 12 \text{ units}$$

$$S = 30 \text{ units}$$

$$T = 20 \text{ units}$$

So, the time taken to complete the whole work alone by,

$$P = (300/15) = 20 \text{ days}$$

$$Q = (300/10) = 30 \text{ days}$$

$$R = (300/12) = 25 \text{ days}$$

$$S = (300/30) = 10 \text{ days}$$

$$T = (300/20) = 15 \text{ days}$$

Answer: D

The total work = 300 units

The combined efficiency of (R+U) = $\{300/(50/3)\} = 18 \text{ units}$

The individual efficiency of R = 12 units

So, the individual efficiency of U = $(18-12) = 6 \text{ units}$

The combined efficiency of (V+U) = $\{300/(300/11)\} = 11 \text{ units}$

So, the individual efficiency of V = $(11-6) = 5 \text{ units}$

Now, the time taken by V to complete the whole work alone = $(300/5) = 60 \text{ days}$.

Therefore, ? = 60.

Hence, the correct answer is option D.

15. Questions

@QuizByMickey

The time taken to complete the whole work by,

$$(P+Q) = 12 \text{ days}$$

$$(R+S+T) = (150/31) \text{ days}$$

$$(P+S) = \{(10/3) \times 2\} = (20/3) \text{ days}$$

$$T = (30/2) = 15 \text{ days}$$

$$(P+R) = (100/9) \text{ days}$$

The total work = 300 units
 (L.C.M. of the numerator
 of the time taken)

So, the efficiency of,

$$(P+Q) = (300/12) = 25 \text{ units}$$

$$(R+S+T) = \{300/(150/31)\} = 62 \text{ units}$$

$$(P+S) = \{300/(20/3)\} = 45 \text{ units}$$

$$T = (300/15) = 20 \text{ units}$$

$$(P+R) = \{300/(100/9)\} = 27 \text{ units}$$

So,

$$(R+S) = (62-20) = 42 \text{ units}$$

$$S = 42 - R$$

$$P + 42 - R = 45 \text{ units}$$

$$P - R = 3 \text{ -----} > (1)$$

$$P + R = 27 \text{ ----} > (2)$$

By solving equations (1) and (2), we get,

$$P = 30/2 = 15$$

$$R = 27 - 15$$

$$R = 12 \text{ units}$$

$$P+Q = 25$$

$$15 + Q = 25$$

$$Q = 10$$

Also,

$$(R+S+T) = 62$$

$$12 + S + 20 = 62$$

$$S = 62 - 32$$

$$S = 30 \text{ units}$$

Now, the individual efficiency of,

@QuizByMickey

$$P = 15 \text{ units}$$

$$Q = 10 \text{ units}$$

$$R = 12 \text{ units}$$

$$S = 30 \text{ units}$$

$$T = 20 \text{ units}$$

So, the time taken to complete the whole work alone by,

$$P = (300/15) = 20 \text{ days}$$

$$Q = (300/10) = 30 \text{ days}$$

$$R = (300/12) = 25 \text{ days}$$

$$S = (300/30) = 10 \text{ days}$$

$$T = (300/20) = 15 \text{ days}$$

Answer: C

The individual efficiency of S = 30 units

The individual efficiency of T = 20 units

Now,

$$\{N*(30+20)\} = \{(2N+2)*20\}$$

$$(N*50) = (40N+40)$$

$$50N = 40N + 40$$

$$10N = 40$$

$$N = 4$$

Therefore, the total work done by S and T = $[\{N*(30+20)\} + \{(2N+2)*20\}] = (50N+40N+40) = (90N+40)$
 $= (90 * 4) + 40 = (360+40) = 400 \text{ units.}$

Hence, the correct answer is option C.

16. Questions

Answer: E

From I,

$$5X^2 - 26X + 32 = 0$$

$$5X^2 - 10X - 16X + 32 = 0$$

$$5X(X-2) - 16(X-2) = 0$$

$$(5X-16)(X-2) = 0$$

$$X = +2 \text{ or } +16/5$$

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From II,

$$3Y^2 - 31Y + 56 = 0$$

$$3Y^2 - 24Y - 7Y + 56 = 0$$

$$3Y(Y-8) - 7(Y-8) = 0$$

$$(3Y-7)(Y-8) = 0$$

$$Y = +7/3, +8$$

X	relation	Y
+2	<	+7/3
+2	<	+8
+16/5	>	+7/3
+16/5	<	+8

Therefore, the relationship cannot be established.

Hence, the correct answer is option E.

17. Questions

Answer: B

From I,

$$4A^2 - 29A + 51 = 0$$

$$4A^2 - 12A - 17A + 51 = 0$$

$$4A(A-3) - 17(A-3) = 0$$

$$(4A-17)(A-3) = 0$$

$$A = +3 \text{ or } +17/4$$

From II,

$$2B^2 - 27B + 90 = 0$$

$$2B^2 - 12B - 15B + 90 = 0$$

$$2B(B-6) - 15(B-6) = 0$$

$$(2B-15)(B-6) = 0$$

$$B = +6 \text{ or } +15/2$$

@QuizByMickey

A	relation	B
+3	<	+6
+3	<	+15/2
+17/4	<	+6
+17/4	<	+15/2

Therefore, $A < B$.

Hence, the correct answer is option B.

18. Questions

Answer: E

From I,

$$P^2 + 4P - 480 = 0$$

$$P^2 + 24P - 20P - 480 = 0$$

$$P(P+24) - 20(P+24) = 0$$

$$(P-20)(P+24) = 0$$

$$P = +20 \text{ or } -24$$

From II,

$$Q^2 - 17Q - 390 = 0$$

$$Q^2 + 13Q - 30Q - 390 = 0$$

$$Q(Q+13) - 30(Q+13) = 0$$

$$(Q-30)(Q+13) = 0$$

$$Q = +30 \text{ or } -13$$

P	Relation	Q
+20	<	+30
+20	>	-13
-24	<	+30
-24	<	-13

Therefore, the relationship cannot be established.

Hence, the correct answer is option E.

19. Questions

Answer: E

@QuizByMickey

From I,

$$M^2 - 42M + 341 = 0$$

$$M^2 - 31M - 11M + 341 = 0$$

$$M(M-31) - 11(M-31) = 0$$

$$(M-11)(M-31) = 0$$

$$M = +11 \text{ or } +31$$

From II,

$$N^2 - 11N - 152 = 0$$

$$N^2 + 8N - 19N - 152 = 0$$

$$N(N+8) - 19(N+8) = 0$$

$$(N-19)(N+8) = 0$$

$$N = +19 \text{ or } -8$$

M	Relation	N
+11	<	+19
+11	>	-8
+31	>	+19
+31	>	-8

Therefore, the relationship cannot be established.

Hence, the correct answer is option E.

20. Questions

Answer: B

From I,

$$6K^2 + 23K - 13 = 0$$

$$6K^2 - 3K + 26K - 13 = 0$$

$$3K(2K - 1) + 13(2K - 1) = 0$$

$$(2K - 1)(3K + 13) = 0$$

$$K = + (1/2) \text{ or } - (13/3)$$

From II,

$$8L^2 - 29L + 21 = 0$$

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$$8L^2 - 8L - 21L + 21 = 0$$

$$8L(L - 1) - 21(L-1) = 0$$

$$(L-1)(8L-21) = 0$$

$$L = +1 \text{ or } +(21/8)$$

K	relation	L
+1/2	<	+1
+1/2	<	+21/8
-13/3	<	+1
-13/3	<	+21/8

Therefore, $K < L$.

Hence, the correct answer is option B.

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